



Artificial intelligence as a merger and acquisition catalyst: A resource-based view[☆]

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ABSTRACT

Mergers and acquisitions (M&A) are pivotal strategic decisions that shape firm evolution. Drawing on the resource-based view, this study uses data from publicly listed firms in China from 2007 to 2024 to examine how artificial intelligence (AI) adoption influences M&A activity. We find that AI adoption significantly increases a firm's likelihood of pursuing M&A, with the effect especially pronounced among firms with abundant cash holdings and those operating in less-developed market environments. The relationship is also stronger for privately owned enterprises, and AI adoption is associated with improved long-term firm value. By linking technological capability with strategic outcomes, this study deepens our understanding of modern M&A drivers, highlighting the critical interplay between internal resources and institutional contexts.

1. Introduction

Mergers and acquisitions (M&A) remains a pivotal strategic instrument for firm renewal and resource reallocation (Beretta et al., 2025; David, 2021; Levine et al., 2020; Masulis et al., 2007). This significance is underscored by 2025 market trends, where global deal values surged 25% year-over-year to exceed \$2 trillion (Mergermarket, 2025; PwC, 2025). Such momentum is particularly pronounced in China; domestic firms announced over \$54 billion in deals during Q1 2025—a 52% annual increase.²

Furthermore, Artificial Intelligence (AI) is fundamentally reshaping the business landscape, profoundly influencing strategic decisions like M&A. While AI's benefits in product development (Cooper & McCausland, 2024) and innovation (Kopka & Fornahl, 2024) are well-established; these analytical strengths are increasingly pivotal in the high-stakes M&A arena. By identifying synergies (Wu & Shang, 2020) and accelerating complex due diligence (Roberts et al., 2022; Rose et al., 2020), AI acts as a general-purpose technology that redefines

competitive advantage (Agrawal et al., 2022; Brynjolfsson & McAfee, 2014). Amid robust market growth (Statista, 2025); M&A is shifting from traditional risk mitigation (Wu et al., 2024) or CSR enhancement (Chen et al., 2023) toward leveraging AI-driven capabilities to secure long-term growth.

Prior studies identify diverse M&A determinants, ranging from leadership traits and rural experiences (Xiao et al., 2024; Zhao et al., 2025) to team stability and ESG alignment (Cai et al., 2025; Tang, Xu, et al., 2024b). Internal governance also plays a key role in resource acquisition and deal efficiency (Xing et al., 2025). Despite this breadth; extant research primarily views digitalization as an acquisition target—examining how firms secure technology through “digital M&A” (e.g.; Tang, Liang, & Fang, 2024a). In contrast; we explore AI adoption as a strategic driver; investigating how internal AI investment enhances a firm's absorptive capacity and motivates subsequent expansion. Given that AI is increasingly pivotal for operational efficiency in China (Liu, Zou, et al., 2025a), this shift from M&A as a means to AI as a motive offers a necessary extension to the literature. Consequently, moving

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² https://pdf.dfcfw.com/pdf/H301_AP202504151656745031_1.pdf

beyond the traditional framework of static financial and governance metrics, this paper contributes to the emerging literature by employing AI adoption as a core explanatory variable, testing whether a firm's endogenous technological capability has emerged as a new strategic driver for M&A decisions.

This study asks: How does a firm's adoption of AI influence its M&A behaviors? Drawing on the resource-based view (RBV), we argue that AI adoption constitutes a dynamic capability—valuable, rare, and inimitable (Barney, 1991). As a complex integration of data; algorithms; and talent (Bharadwaj, 2000; Petcu et al., 2024), AI generates firm-level heterogeneity that is difficult to replicate (Wang et al., 2023). Firms thus use AI-driven insights to identify and acquire complementary assets. However, the effectiveness of AI depends on both internal and external conditions. Internally, cash flow serves as an enabling resource for capital-intensive deals. Externally, the level of marketization negatively moderates this relationship, as higher information efficiency in developed markets may diminish the relative advantage of using M&A to capitalize on an AI-driven informational edge.

Using data from Chinese listed firms, this study makes several contributions. First, we extend the RBV by conceptualizing AI as a contingent dynamic capability (Kraaijenbrink et al., 2010). While prior research emphasizes traditional assets or executive traits (Guizani & Larabi, 2025; Xiao et al., 2024), we suggest that the strategic role of AI depends on the interplay between internal financial resources and external institutional environments (Zhang et al., 2025). Second, we broaden M&A literature by shifting focus from market power (Bonaime & Wang, 2024) to technology-based strategic capabilities (Liang et al., 2024). By reversing the typical causality—modeling AI as a driver rather than an outcome—we complement work on digital M&A and innovation (Tang, Liang, & Fang, 2024a) and technological diversification (Arts et al., 2025; Zhou, Liang, et al., 2025a). Finally, we deepen the resource orchestration literature by showing how AI and financial resources jointly enable M&A (Li et al., 2024).

2. Literature review and research hypothesis

2.1. RBV and AI adoption

This paper employs the Resource-Based View (RBV) to explain how AI adoption shapes M&A strategies (Estensoro et al., 2022; Lockett et al., 2009). The RBV posits that sustainable competitive advantage stems from unique bundles of valuable, scarce, inimitable, and non-substitutable resources (Lockett et al., 2009); provided these assets are managed effectively (Estensoro et al., 2022; Sirmon et al., 2011). As a transformative asset (Elia et al., 2021), AI adoption is a strategic initiative tied to firm-specific attributes (Proksch et al., 2024), creating capabilities that are difficult for rivals to replicate.

Recent literature applies the RBV to analyze AI as a critical organizational asset. Krakowski et al. (2023) link competitive advantage to unique resource endowments; highlighting the substitution and complementation dynamics generated by AI. Building on this, Fosso Wamba et al. (2024) categorized AI assets into three types to investigate their synergistic impacts on firm performance. Furthermore, Li, Zhang, and Um (2025b) frame AI as a crucial resource for advancing positive ESG outcomes.

While early scholarship recognized information technology as a strategic asset (Bharadwaj, 2000; Lioukas et al., 2016), AI differs from conventional technologies like IoT by integrating vast datasets with machine learning to detect latent patterns. Its predictive strength generates new tacit knowledge (Prior & Marcos-Cuevas, 2025); such as personalized services based on historical customer data (Liu, Wang, & Zhu, 2025b). This deep operational embedding makes AI-driven advantages difficult to imitate (Lichtenthaler, 2019). Consequently, AI capabilities fit squarely within the RBV (Petcu et al., 2024). Beyond mere deployment, AI adoption necessitates reconfiguring operational routines and decision-making systems (Mishra et al., 2022), creating a

forward-looking value proposition that reshapes the firm's performance.

2.2. Influence of AI adoption on M&A

Firms frequently operate under resource constraints (Kalatzis et al., 2025); motivating them to leverage AI to enhance their resource portfolios (Ren et al., 2024). Grounded in the RBV, we argue that AI adoption provides distinct advantages that drive M&A behaviors through three primary mechanisms: reduced labor costs, improved management efficiency, and enhanced knowledge spillovers within industrial clusters.

First, AI improves operational efficiency by automating core processes, such as manufacturing and logistics, thereby substituting manual labor and reducing costs (Acemoglu & Restrepo, 2020). These savings expand available capital; increasing the firm's capacity for M&A (Fan & Yang, 2025; Neary, 2007). Second, AI optimizes organizational management by overcoming traditional information bottlenecks and coordination difficulties (Haefner et al., 2021). By integrating large-scale data in real time; AI enhances decision accuracy and lowers managerial costs (Cho & Lee, 2019; Li, Wu, et al., 2025a), allowing firms to better coordinate the acquisition of strategic resources (Kushermanto & Rohman, 2024). Third, AI strengthens absorptive capacity; enabling firms to more effectively convert external knowledge spillovers into proprietary innovation (Cohen & Levinthal, 1990; Rezaei et al., 2024). Consequently, AI-driven innovation engines motivate firms to pursue acquisitions of complementary technologies or teams. Based on these mechanisms, we propose the following hypothesis:

H1. AI adoption is positively associated with firm M&A.

2.3. The moderating role of marketization

Marketization reflects the degree to which resource allocation relies on market mechanisms over state planning (Du et al., 2017; Yu et al., 2025). Measured across five dimensions—including government-market relations, nonstate economy development, and legal sophistication—the marketization index (Fan et al., 2001) captures the maturity of the institutional environment. According to the “institutional substitution” hypothesis; well-functioning markets provide effective substitutes for a firm's internal capabilities (Nayyar & Luiz, 2024). In highly marketized regions; firms can readily purchase professional due diligence; industry analysis; and valuation models from external providers (Wang et al., 2025). These services replicate core pre-M&A functions of internal AI; such as target screening (Song et al., 2024), thereby eroding the rarity and non-substitutability of a firm's proprietary technology. As competitors gain access to similar decision-support through market channels, the asymmetric information advantage of internal AI diminishes, diluting its unique strategic value and marginal contribution to M&A.

Institutional theory suggests that highly marketized environments increase isomorphic pressures, compelling firms to conform to established decision-making norms (Liu, Zhou, & Wu, 2024a). Because M&A transactions are high-risk and highly visible; managers in mature markets often avoid unconventional opportunities identified by AI—such as high-potential but low-profit startups—to mitigate reputational or valuation risks (Grashof & Kopka, 2023; Krause et al., 2019; Rao et al., 2023). Consequently, even when AI reveals strategically valuable but non-consensus targets, the need to preserve organizational legitimacy may lead managers to reject these insights. This suggests that advanced institutional environments may hinder the translation of AI capabilities into actual M&A choices. Based on this logic, we propose:

H2. Marketization negatively moderates the relationship between AI adoption and firm M&A, weakening the positive effect of AI on M&A in highly marketized regions.

Table 1
Variable definitions.

Variable Type	Variable Symbol	Variable Definition
Dependent	<i>M&A</i>	Ln [1 + M&A transaction value]
Independent (AI adoption)	<i>AI</i>	Ln [1 + Frequency of AI-related terms mentioned in annual report]
Moderators	<i>Market</i>	Fan Gang Marketization Index of firm
	<i>Cashhold</i>	(Cash and cash equivalents + Net short-term investments) / Total assets
	<i>FirmAge</i>	Ln [#Years since firm establishment]
	<i>Size</i>	Ln [End-of-FY – total assets]
	<i>Lev</i>	Total liabilities / Total assets (proxy for firm leverage)
	<i>Fixed</i>	Fixed assets / Total assets
Controls	<i>Agent</i>	Administrative expenses/Operating revenue (proxy for agency costs)
	<i>Indep</i>	#Independent directors / #Directors
	<i>Dual</i>	= 1 if Chairman is General manager; otherwise 0
	<i>Big4</i>	= 1 if Firm audited by one of the Big Four accounting firms; otherwise 0

Table 2
Descriptive statistical analysis.

Variable	Mean	SD	Minimum	Maximum
<i>M&A</i>	4.6747	8.1289	0.0000	22.1128
<i>AI</i>	0.8980	1.2409	0.0000	6.4968
<i>Market</i>	9.5257	1.6506	0.1180	11.4940
<i>Cashhold</i>	0.1943	0.1394	0.0005	0.8574
<i>FirmAge</i>	2.9052	0.3707	1.0986	3.6636
<i>Size</i>	22.1706	1.2952	19.4058	26.4523
<i>Lev</i>	0.4125	0.2048	0.0274	0.9347
<i>Fixed</i>	0.2124	0.1579	0.0016	0.7694
<i>Agent</i>	0.3922	2.3412	0.0057	39.1655
<i>Indep</i>	0.3753	0.0532	0.2500	0.6000
<i>Dual</i>	0.2923	0.4548	0.0000	1.0000
<i>Big4</i>	0.0627	0.2424	0.0000	1.0000

Note: *N* = 47,509.

2.4. Moderating role of firm's cash holdings

Cash holdings, defined as liquid assets including marketable securities and unused debt capacity (Opler et al., 2001); represent fungible slack resources (Martin & Cuypers, 2024). These resources provide maximum flexibility for navigating uncertainty (Kim & Bettis, 2014;

Natividad, 2013; O'Brien & Folta, 2009). From the RBV, financial slack affords crucial managerial discretion (George, 2005) but must be integrated with other assets to be effective. For instance; the strategic value of organizational learning—whether direct or vicarious—depends significantly on available cash (Martin & Cuypers, 2024).

On the one hand, cash holdings confer competitive advantages in dynamic environments (Lyandres & Palazzo, 2016). While AI identifies high-value M&A opportunities; successful execution requires strategic flexibility (Kim & Bettis, 2014). Internal reserves allow firms to act immediately without the costs and information asymmetries associated with external financing (Hennessy & Whited, 2007). Avoiding external capital also prevents premature disclosure of strategic intent; which could alert competitors or inflate prices (Bhattacharya & Ritter, 1983). Furthermore; cash-heavy bids enhance bargaining power and deal attractiveness; facilitating favorable terms and reducing failure risks (Vo et al., 2024).

Conversely, AI provides the foresight and agility (Wang et al., 2022) to identify unconventional targets; such as innovative firms with disruptive technologies that may lack short-term profitability (Haefner et al., 2021). While external investors may view such radical innovations conservatively; large cash reserves act as a “risk capital buffer” (Grashof & Kopka, 2023). This financial cushion enables firms to internally finance and absorb the risks of pursuing non-consensus targets; allowing them to convert AI-driven insights into long-term strategic value (Lyandres & Palazzo, 2016; Shah et al., 2025). Therefore, we propose the following hypothesis:

H3. Cash holdings positively moderate the relationship between AI adoption and firm M&A. The positive effect of AI adoption on a firm's M&A is stronger when its cash holdings are higher.

3. Research design

3.1. Sample selection

To empirically examine the association between AI adoption and M&A behavior, this study selects all A-share listed firms on the Shanghai and Shenzhen stock exchanges in China from 2007 to 2024 as the research sample. The year 2007 marks the full implementation of the revised Chinese Accounting Standards, which significantly enhanced the standardization and comparability of firm information disclosure, thereby providing institutional support for AI-related textual analysis. Regarding data sources, the textual content of annual reports is obtained

Table 3
Correlation analysis.

Variables	(1)	(2)	(3)	(4)	(5)	(6)
(1) <i>M&A</i>	1.0000					
(2) <i>AI</i>	−0.0399***	1.0000				
(3) <i>Market</i>	−0.0644***	0.2543***	1.0000			
(4) <i>Cashhold</i>	−0.0556***	0.1164***	0.0579***	1.0000		
(5) <i>FirmAge</i>	−0.0797***	0.1145***	0.1713***	−0.1767***	1.0000	
(6) <i>Size</i>	0.0622***	0.0118***	−0.0256***	−0.2433***	0.2168***	1.0000
(7) <i>Lev</i>	0.0769***	−0.1036***	−0.1248***	−0.4010***	0.1318***	0.4829***
(8) <i>Fixed</i>	0.0015	−0.2836***	−0.2061***	−0.3538***	−0.0105**	0.1167***
(9) <i>Agent</i>	0.0058	−0.0168***	−0.0114**	−0.0093**	0.0648***	0.0579***
(10) <i>Indep</i>	−0.0120***	0.0788***	0.0670***	−0.0003	0.0300***	0.0221***
(11) <i>Dual</i>	−0.0293***	0.1347***	0.1693***	0.1028***	−0.0593***	−0.1753***
(12) <i>Big4</i>	0.0023	−0.0079*	0.0370***	−0.0385***	0.0116**	0.3361***
(7) <i>Lev</i>	1.0000		(9)	(10)	(11)	(12)
(8) <i>Fixed</i>	0.1334***	1.0000				
(9) <i>Agent</i>	0.0298***	−0.0243***	1.0000			
(10) <i>Indep</i>	−0.0195***	−0.0483***	0.0006	1.0000		
(11) <i>Dual</i>	−0.1615***	−0.1101***	−0.0374***	0.1040***	1.0000	
(12) <i>Big4</i>	0.0947***	0.0359***	0.0205***	0.0352***	−0.0551***	1.0000

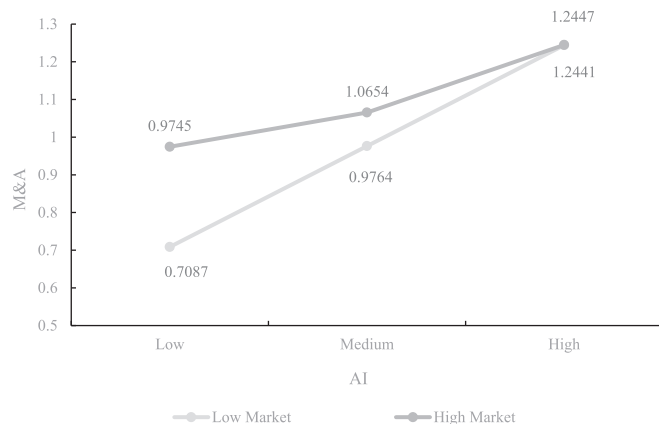
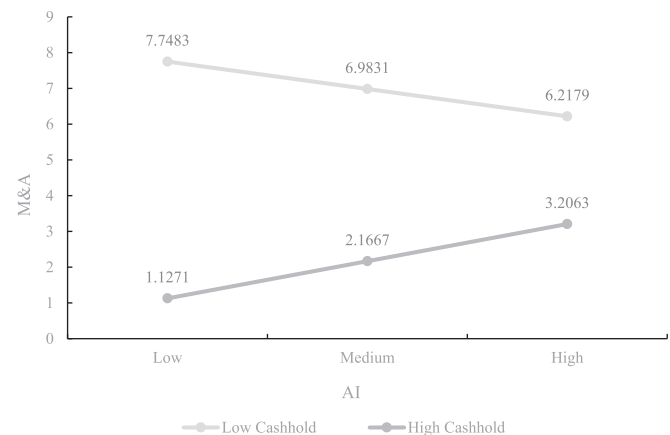
Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively. *N* = 47,509.

Table 4

Results of the main regression test.

	Model 1	Model 2	Model 3	Model 4	Model 5
	DV = M&A	DV = M&A	DV = M&A	DV = M&A	DV = M&A
AI	0.2271*** (0.0706)	0.1606** (0.0714)	0.2004*** (0.0743)	0.1372* (0.0720)	0.1793** (0.0747)
Market			0.0326 (0.1348)		0.0445 (0.1349)
AI×Market			−0.0824** (0.0418)		−0.0884** (0.0418)
Cashhold				−2.4082*** (0.4450)	−2.4572*** (0.4452)
AI×Cashhold				0.9024*** (0.2999)	0.8846*** (0.2998)
FirmAge		−0.1305 (0.5477)	−0.1058 (0.5475)	−0.6161 (0.5572)	−0.5951 (0.5569)
Size		0.4689*** (0.1075)	0.4722*** (0.1075)	0.4705*** (0.1077)	0.4739*** (0.1077)
Lev		0.7614* (0.4248)	0.7990* (0.4258)	0.1720 (0.4330)	0.2018 (0.4336)
Fixed		−1.7926*** (0.5639)	−1.7675*** (0.5642)	−2.8063*** (0.5926)	−2.7942*** (0.5927)
Agent		−0.0207 (0.0192)	−0.0210 (0.0192)	−0.0203 (0.0192)	−0.0205 (0.0192)
Indep		0.9828 (1.1406)	0.9791 (1.1411)	0.8609 (1.1409)	0.8547 (1.1414)
Dual		−0.0069 (0.1376)	−0.0087 (0.1375)	0.0097 (0.1375)	0.0080 (0.1374)
Big4		−0.1041 (0.3410)	−0.1081 (0.3401)	−0.1170 (0.3423)	−0.1209 (0.3414)
Firm fixed effect	YES	YES	YES	YES	YES
Year fixed effect	YES	YES	YES	YES	YES
Industry fixed effect	YES	YES	YES	YES	YES
Constant	4.4708*** (0.0634)	−5.7715** (2.7059)	−6.2392** (3.0098)	−3.4251 (2.7397)	−3.9791 (3.0406)
N	47,509	47,509	47,509	47,509	47,509
R ² _adjusted	0.0985	0.0995	0.0996	0.0996	0.0996

Note: ***, **, and * represent significance levels of 1%, 5%, and 10% respectively. Values in parentheses are standard errors.

**Fig. 1.** Moderating effect of marketization.**Fig. 2.** Moderating effect of cash holdings.

from CNINFO (<https://www.cninfo.com.cn/>), which is used to extract firm-level information on AI technology application. Data on M&A activities, financial indicators, and firm governance variables are sourced from the China Stock Market and Accounting Research (CSMAR) database.

The initial sample is refined as follows: (1) financial firms, including banks, securities companies, and insurance institutions, are excluded; (2) firms labeled with ST, *ST, or PT, which indicate abnormal business operations, are removed; (3) observations with substantial missing data are excluded. After this screening process, the final sample comprises 47,509 firm-year observations from 4904 listed firms.

3.2. Variable definitions

3.2.1. Dependent variable: M&A behavior (M&A)

Following Xiao et al. (2024), we measure M&A behavior using the annual transaction value of acquisitions by listed firms, with data sourced from the CSMAR M&A Database (2007–2024). To ensure the sample reflects substantive strategic activities rather than internal adjustments, we apply several filters: (1) excluding transactions below RMB 1 million; (2) removing related-party transactions to avoid capturing internal capital reallocation; and (3) excluding restructuring events such as asset swaps, debt restructuring, share repurchases, and tender offers. After aggregating these activities by acquirer and fiscal

Table 5
Instrumental variable regression results.

	Model 6	Model 7
	DV = AI	DV = M&A
AI		0.3366** (0.1430)
AI – iv	0.3226*** (0.0068)	
FirmAge	0.0931 (0.0592)	–0.1805 (0.5492)
Size	0.1380*** (0.0113)	0.4303*** (0.1110)
Lev	–0.0227 (0.0385)	0.7496* (0.4247)
Fixed	–0.2434*** (0.0470)	–1.7621*** (0.5636)
Agent	0.0009 (0.0013)	–0.0209 (0.0192)
Indep	–0.3968*** (0.0940)	1.0422 (1.1399)
Dual	0.0010 (0.0123)	–0.0063 (0.1378)
Big4	–0.0449 (0.0335)	–0.0928 (0.3412)
Firm fixed effect	YES	YES
Year fixed effect	YES	YES
Industry fixed effect	YES	YES
Kleibergen – Paap rk LM	784.720***	
Kleibergen – Paap rk Wald F	2246.144	
N	47,509	47,509
R ² adjusted	0.8247	

Note: ***, **, and * represent significance levels of 1%, 5%, and 10% respectively. Values in parentheses are standard errors.

year, the dependent variable (M&A) is defined as the natural logarithm of the total transaction value plus one, which mitigates distributional skewness while preserving observations with zero M&A activity.

Table 6
Heckman two–step estimation.

	Model 8	Model 9
	DV = M&A dummy	DV = M&A
AI	0.0331*** (0.0066)	2.0293*** (0.2496)
Oversea	0.1420*** (0.0522)	7.9665*** (1.0887)
IMR		74.0815*** (9.4450)
FirmAge	–0.0726*** (0.0213)	–3.9904*** (0.7468)
Size	0.0613*** (0.0065)	3.9172*** (0.4589)
Lev	0.2381*** (0.0387)	13.8555*** (1.7379)
Fixed	–0.2563*** (0.0488)	–15.9264*** (1.8950)
Agent	–0.0079*** (0.0027)	–0.4620*** (0.0589)
Indep	0.0616 (0.1215)	4.3846*** (1.2252)
Dual	0.0134 (0.0149)	0.7637*** (0.1659)
Big4	–0.1790*** (0.0287)	–10.1790*** (1.3530)
Firm fixed effect	NO	YES
Year fixed effect	YES	YES
Industry fixed effect	YES	YES
Constant	–1.5410*** (0.1596)	–170*** (21.5015)
N	47,507	47,507
R ² adjusted		0.1013

Note: ***, **, and * represent significance levels of 1%, 5%, and 10% respectively. Values in parentheses are standard errors.

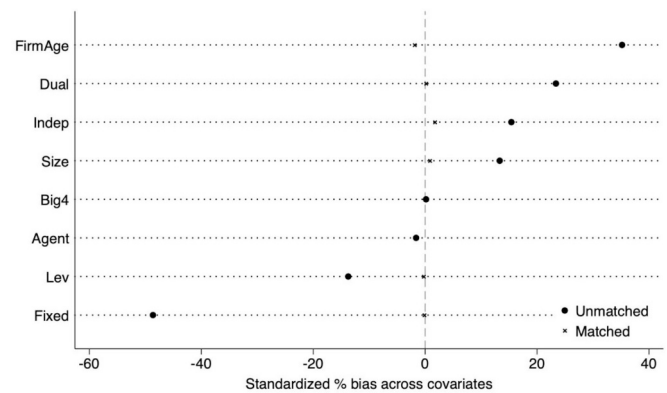


Fig. 3. Balance test.

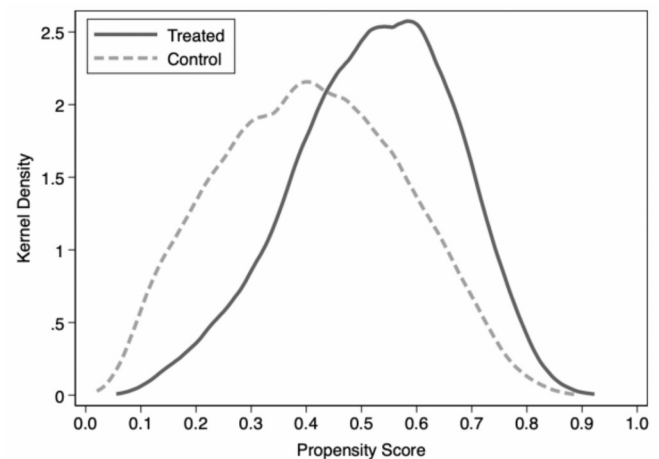


Fig. 4. Kernel density plot before matching.

3.2.2. Independent variable: AI adoption (AI)

Following Yao et al. (2024) and Mishra et al. (2022); we constructed a firm-level AI adoption indicator via textual analysis of annual reports from the CNINFO platform. We first processed these reports using the “jieba” Python library for Chinese word segmentation. Next; we integrated a validated 73-term AI lexicon from Yao et al. (2024)—including keywords such as “artificial intelligence” and “machine learning”—as a

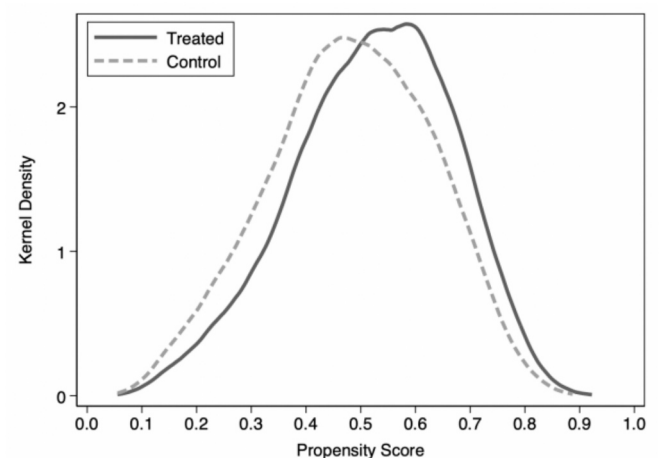


Fig. 5. Kernel density plot after matching.

Table 7
Propensity score matching (PSM) results.

	Model 10	Model 11
	DV = M&A	DV = M&A
AI	0.2198** (0.0964)	0.1775* (0.0985)
FirmAge		-1.0207 (0.9003)
Size		0.6500*** (0.1636)
Lev		0.2678 (0.6516)
Fixed		-1.5292* (0.8769)
Agent		0.0065 (0.0301)
Indep		0.0136 (0.0169)
Dual		0.0790 (0.2008)
Big4		-0.4557 (0.5191)
Firm fixed effect	YES	YES
Year fixed effect	YES	YES
Industry fixed effect	YES	YES
Constant	4.4793*** (0.0890)	-7.2316* (4.1150)
N	22,584	22,584
R ² adjusted	0.1069	0.1078

Note: ***, **, and * represent significance levels of 1%, 5%, and 10% respectively. Values in parentheses are standard errors.

custom dictionary to calculate the frequency of AI-related terms (see Appendix). To mitigate skewness, the final AI adoption measure is the natural logarithm of the aggregated keyword frequency plus one.

3.2.3. Moderator: marketization (market)

Following Gao (2025) and Wang et al. (2021); we measure regional marketization using the Fan Gang Marketization Index from the NERI Report. This index aggregates five dimensions: government-market relations; non-state economy development; and the maturity of product markets; factor markets; and legal/intermediary environments. To extend the dataset beyond 2021; we follow Xie and Zhu (2021) by applying the average historical growth rate of each region's total index to estimate subsequent values.

3.2.4. Moderator: cash holdings (cashhold)

In line with Nnadi et al. (2022), cash holdings are defined as the ratio

of monetary funds plus net short-term investments to total assets.

3.2.5. Controls

Following prior literature (Xiao et al., 2024), we include a set of firm-level control variables to account for heterogeneity in firm characteristics. Specifically, we control for firm age (*FirmAge*), firm size (*Size*), leverage ratio (*Lev*), the proportion of fixed assets (*Fixed*), agency costs (*Agent*), the proportion of independent directors on the board (*Indep*), whether the chairman also serves as the general manager (*Dual*), and whether the firm is audited by one of the Big Four accounting firms (*Big4*). Detailed definitions and measurement methods for all variables are presented in Table 1.

3.3. Model construction

Drawing on the preceding theoretical analysis and research hypotheses, this study develops a panel data regression model to empirically examine the relationship between firm AI adoption and M&A behavior.

$$M\&A_{i,t} = \beta_0 + \beta_1 AI_{i,t} + \beta_2 FirmAge_{i,t} + \beta_3 Size_{i,t} + \beta_4 Lev_{i,t} + \beta_5 Fixed_{i,t} + \beta_6 Agent_{i,t} + \beta_7 Indep_{i,t}$$

$$+ \beta_8 Dual_{i,t} + \beta_9 Big4_{i,t} + \beta_{10} Market_{i,t} + \beta_{11} Cashhold_{i,t} +$$

$$\beta_{12} AI_{i,t} \times Market_{i,t} + \beta_{13} AI_{i,t} \times Cashhold_{i,t} + \lambda_i + \mu_t + \delta_r + \varepsilon_{i,t} \quad (1)$$

i , t and r in Eq. (1) denote the firm, year and industry respectively. We incorporate firm fixed effects, year fixed effects, and industry fixed effects to account for time-invariant unobserved heterogeneity at the firm level, macro-level shocks over time, and industry-specific influences, thereby minimizing potential estimation bias. λ_i , μ_t and δ_r represent firm, year, and industry fixed effects, respectively. $\varepsilon_{i,t}$ is the idiosyncratic error term, and standard errors are clustered at the firm level to address potential autocorrelation and heteroskedasticity. To reduce the influence of extreme outliers, all key continuous variables are winsorized at the 1st and 99th percentiles. Additionally, to alleviate potential multicollinearity, relevant continuous variables are mean-centered before constructing interaction terms.

4. Empirical analysis

4.1. Descriptive statistical analysis and correlation analysis

Table 2 presents the descriptive statistics. *M&A* (mean = 4.6747; SD

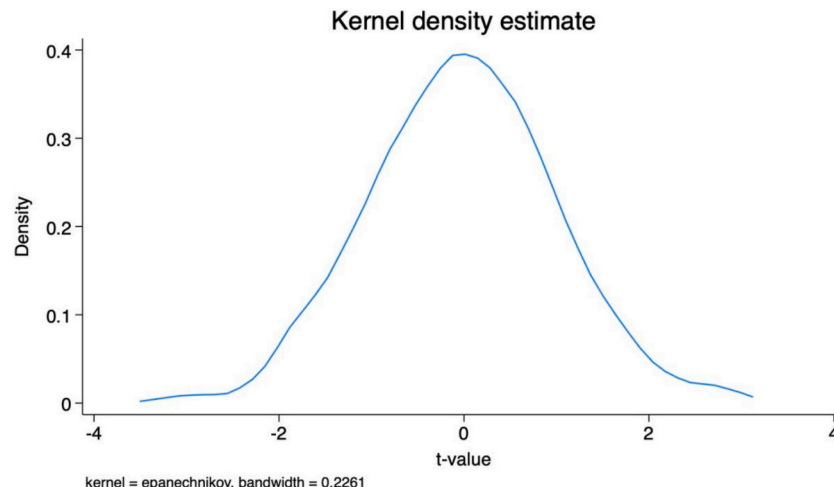


Fig. 6. Placebo test.

Table 8

Generalized method of moments (GMM) dynamic panel analysis.

	Model 12
	DV = M&A
L.M&A	0.02123** (0.01034)
AI	0.8024** (0.3810)
FirmAge	−10.7431** (5.2359)
Size	−0.7189 (1.5058)
Lev	6.0731 (4.3169)
Fixed	14.3532 (12.7415)
Agent	0.5548 (1.0082)
Indep	−0.4685 (1.8037)
Dual	−0.3516 (0.5652)
Big4	0.7116 (2.1669)
Year fixed effect	YES
N	41,520
AR(1)	0.0000
AR(2)	0.1220
Hansen	0.1890

Note: ***, **, and * represent significance levels of 1%, 5%, and 10% respectively. Values in parentheses are standard errors.

= 8.1289) ranges from 0 to 22.1128, indicating substantial variation in firm behavior. AI (mean = 0.8980; SD = 1.2409) similarly reflects significant differences in adoption across the sample. *Market* and *Cashhold* show mean values of 9.5257 and 0.1943, respectively, with moderate dispersion. Control variables—including *FirmAge*, *Size*, *Lev*, and *Fixed*—display reasonable variation in firm scale, leverage, and asset structure. Notably, *Dual* (mean = 0.2923) suggests that approximately 29% of firms exhibit CEO-chair duality, while *Big4* (mean = 0.0627) indicates that only a small proportion of the sample is audited by Big Four accounting firms.

Table 3 presents the correlation matrix of the main variables. Most correlation coefficients have absolute values below 0.5, indicating that multicollinearity is not a major concern in the model. To further explore the relationship between AI adoption and M&A behavior, regression analyses are conducted in the following sections.

4.2. Benchmark regression

Before the formal regression, multicollinearity was assessed using the Variance Inflation Factor (VIF). Results show a maximum VIF of 1.55 and an average of 1.22—well below the conventional threshold of 10—indicating that multicollinearity is not a concern in the model.

4.2.1. The association between AI adoption and M&A behavior

Table 4 presents the regression results examining the the association between AI adoption and M&A behavior. Model 1, which includes firm, year, and industry fixed effects, shows a positive and statistically significant coefficient for AI ($\beta = 0.2271$, $p < 0.01$). In Model 2, after introducing all control variables, the coefficient remains robust and significant ($\beta = 0.1606$, $p < 0.05$), providing empirical support for H1. Regarding controls, firm size and financial leverage are positively

Table 9

Changing the measurement of the independent variable.

	Model 13	Model 14	Model 15	Model 16
	DV = M&A	DV = M&A	DV = M&A	DV = M&A
AI_MD&A	0.1110* (0.0633)	0.1456** (0.0681)	0.0843 (0.0639)	0.1225* (0.0685)
Market		0.0464 (0.1338)		0.0576 (0.1339)
AI × Market		−0.0618 (0.0412)		−0.0685* (0.0412)
Cashhold			−2.5257*** (0.4420)	−2.5563*** (0.4414)
AI × Cashhold			0.5737* (0.2997)	0.5679* (0.2993)
FirmAge	−0.0962 (0.5475)	−0.0858 (0.5473)	−0.5406 (0.5566)	−0.5335 (0.5564)
Size	0.4854*** (0.1072)	0.4857*** (0.1072)	0.4822*** (0.1074)	0.4825*** (0.1073)
Lev	0.7697* (0.4247)	0.7867* (0.4252)	0.1837 (0.4333)	0.1956 (0.4337)
Fixed	−1.8016*** (0.5636)	−1.7864*** (0.5636)	−2.7849*** (0.5916)	−2.7783*** (0.5916)
Agent	−0.0205 (0.0192)	−0.0206 (0.0192)	−0.0200 (0.0192)	−0.0201 (0.0192)
Indep	0.9645 (1.1407)	0.9731 (1.1407)	0.8313 (1.1420)	0.8396 (1.1419)
Dual	−0.0057 (0.1376)	−0.0081 (0.1376)	0.0103 (0.1375)	0.0078 (0.1375)
Big4	−0.1098 (0.3408)	−0.1119 (0.3402)	−0.1160 (0.3418)	−0.1181 (0.3412)
Firm fixed effect	YES	YES	YES	YES
Year fixed effect	YES	YES	YES	YES
Industry fixed effect	YES	YES	YES	YES
Constant	−6.1588** (2.6979)	−6.6511** (3.0044)	−3.8040 (2.7349)	−4.3817 (3.0375)
N	47,509	47,509	47,509	47,509
R ² _adjusted	0.0994	0.0994	0.1002	0.1002

Note: ***, **, and * represent significance levels of 1%, 5%, and 10% respectively. Values in parentheses are standard errors.

Table 10
Changing the measurement of the independent variable.

	Model 17	Model 18	Model 19
	DV = M&A	DV = M&A	DV = M&A
<i>Allevel</i>	16.2215*** (4.1677)	17.3563*** (4.1081)	17.2652*** (4.1069)
<i>Market</i>			0.0711 (0.1336)
<i>AI × Market</i>			−1.5672 (2.6171)
<i>Cashhold</i>		−2.5279*** (0.4404)	−2.5378*** (0.4402)
<i>AI × Cashhold</i>		49.7534** (25.3549)	49.1473* (25.5865)
<i>FirmAge</i>	−0.0607 (0.5475)	−0.4793 (0.5561)	−0.4750 (0.5561)
<i>Size</i>	0.4966*** (0.1068)	0.4909*** (0.1070)	0.4909*** (0.1070)
<i>Lev</i>	0.8052* (0.4241)	0.2301 (0.4326)	0.2281 (0.4329)
<i>Fixed</i>	−1.9640*** (0.5646)	−2.8928*** (0.5895)	−2.8970*** (0.5897)
<i>Agent</i>	−0.0202 (0.0191)	−0.0203 (0.0190)	−0.0204 (0.0190)
<i>Indep</i>	0.9449 (1.1395)	0.8734 (1.1403)	0.8740 (1.1404)
<i>Dual</i>	−0.0123 (0.1374)	−0.0016 (0.1374)	0.0023 (0.1374)
<i>Big4</i>	−0.1250 (0.3414)	−0.1253 (0.3424)	−0.1230 (0.3424)
<i>Firm fixed effect</i>	YES	YES	YES
<i>Year fixed effect</i>	YES	YES	YES
<i>Industry fixed effect</i>	YES	YES	YES
<i>Constant</i>	−6.4749** (2.6943)	−4.1925 (2.7336)	−4.8757 (3.0393)
<i>N</i>	47,509	47,509	47,509
<i>R²_{adjusted}</i>	0.0997	0.1005	0.1005

Note: ***, **, and * represent significance levels of 1%, 5%, and 10% respectively. Values in parentheses are standard errors.

associated with M&A, consistent with [Xiao et al. \(2024\)](#), who note that larger and more leveraged firms frequently pursue strategic acquisitions. Conversely, fixed assets exhibit a negative relationship with M&A, likely because substantial physical assets increase the difficulty of resource reallocation during restructuring. Overall, these findings confirm that AI adoption is positively associated with firm M&A activity.

4.2.2. The moderating role of marketization

[Table 4](#) also presents the results of the moderation analysis. In Model 3, the interaction between *AI* and *Market* (*AI*×*Market*) is introduced to examine the moderating effect of regional marketization. Model 4 replaces *Market* with *Cashhold* to test the moderating role of firm-level cash holdings. Model 5 incorporates both moderators and their respective interaction terms.

The coefficients of interaction term *AI*×*Market* are significantly negative in both Model 3 and Model 5 ($\beta = -0.0824$, $p < 0.05$; $\beta = -0.0884$, $p < 0.05$), indicating that the positive effect of *AI* is weaker in regions with higher levels of marketization. These findings provide empirical support for H2. The plot of the marginal effects of marketization is shown in [Fig. 1](#).

4.2.3. The moderating role of cash holdings

From [Table 4](#), we can see that the coefficients of the interaction between *AI* and *Cashhold* (*AI*×*Cashhold*) in Model 4 and Model 5 are significantly positive ($\beta = 0.9024$, $p < 0.01$; $\beta = 0.8846$, $p < 0.01$), indicating that the effect of *AI* on M&A is significantly strengthened when cash holdings increase. These results provide empirical support for H3. The plot of the marginal effects of cash holdings is shown in [Fig. 2](#).

Besides, the coefficient of *AI* remains positive and statistically significant across three models ($\beta = 0.2004$, $p < 0.01$; $\beta = 0.1372$, $p < 0.1$;

$\beta = 0.1793$, $p < 0.05$), consistently validating H1.

4.3. Endogeneity issues and tests

4.3.1. Instrumental variable regression

Following [Goldsmith-Pinkham et al. \(2020\)](#), we construct an instrumental variable (*AI_{iv}*) using the shift-share (share-move) method to address endogeneity. Specifically, we multiply each firm's lagged AI adoption level by the sample-wide annual growth rate of AI adoption, excluding the focal firm from the growth calculation to ensure the instrument reflects aggregate technological evolution rather than firm-specific shocks. This approach simulates firm-level AI growth based on historical exposure to aggregate trends, providing a cleaner identification of the association between AI and M&A behavior. [Table 5](#) reports the two-stage least squares (2SLS) estimation results.

In Model 6, *AI* is significantly associated with the instrumental variable at the 1% significance level, indicating that the instrumental variable has strong explanatory power. The *Kleibergen-Paap rk LM* statistic is 784.720, which significantly rejects the null hypothesis of under-identification at the 1% level, suggesting that the model is not under-identified. Furthermore, the *Kleibergen-Paap rk Wald F* statistic reaches 2246.144, which far exceeds the *Stock-Yogo* critical threshold of 16.38 for weak instrument diagnosis, thereby ruling out concerns about instrument weakness. In Model 7, the coefficient of *AI* remains significantly positive at the 5% level ($\beta = 0.3366$, $p < 0.05$), providing further empirical support for the argument that AI adoption promotes M&A behavior. In addition, we conduct the *Durbin-Wu-Hausman* test on *AI*. The results show that the *Hausman* statistic is 7.7480 and the *p-value* is 0.0054, indicating that there is an endogeneity issue with the independent variable *AI* in the benchmark regression model.

4.3.2. Heckman two-step estimation

To address potential self-selection bias, we employ the Heckman two-step method. In the first step, a Probit model estimates the likelihood of M&A participation using executive overseas experience (*Oversea*) as the identifying variable. Prior research indicates that international exposure broadens managerial cognition and significantly increases the propensity to initiate M&A decisions ([Agcayazi et al., 2024](#); [Ding et al., 2021](#); [Liu & Kang, 2025](#)). As shown in Model 8 of [Table 6](#), *AI* and *Oversea* are both positively and significantly associated with the likelihood of M&A behavior ($\beta = 0.0331$, $p < 0.01$; $\beta = 0.1420$, $p < 0.01$), which is consistent with the relevance of the selection equation.

Based on the first-stage estimation, the inverse Mills ratio (IMR) is computed and introduced into the second-stage outcome equation to correct for selection bias. In Model 9, the coefficient of *IMR* is significantly positive ($\beta = 74.0815$, $p < 0.01$), suggesting that unobserved factors affecting selection into M&A are also correlated with firm's M&A behavior. After controlling for sample selection, the regression coefficient of *AI* remains positive and significant ($\beta = 2.0293$, $p < 0.01$). The above results are consistent with the robustness of the findings after accounting for potential selection bias.

4.3.3. Propensity score matching

To address endogeneity stemming from the non-random nature of AI adoption, we employ Propensity Score Matching (PSM). Firms are divided into treatment and control groups based on the presence of AI-related keywords. We perform 1:1 nearest-neighbor matching with replacement using the previously defined control variables. As illustrated in [Fig. 3](#), the balance test confirms that the matching procedure effectively mitigates selection bias: pre-matching covariate imbalances are significantly reduced, with post-matching standardized differences falling below 10% and *t*-tests showing no significant differences between the groups. These results ensure that systematic differences in observable characteristics are minimized for the empirical analysis.

[Figs. 4 and 5](#) illustrate the kernel density distributions for the

Table 11
Changing the measurement of the dependent variable.

	Model 20	Model 21	Model 22	Model 23
	DV = <i>M&A_1</i>	DV = <i>M&A_1</i>	DV = <i>M&A_1</i>	DV = <i>M&A_1</i>
<i>AI</i>	0.1663** (0.0740)	0.2075*** (0.0770)	0.1421* (0.0747)	0.1856** (0.0774)
<i>Market</i>		0.0332 (0.1397)		0.0455 (0.1398)
<i>AI × Market</i>		−0.0852** (0.0433)		−0.0915** (0.0434)
<i>Cashhold</i>			−2.4950*** (0.4617)	−2.5457*** (0.4618)
<i>AI × Cashhold</i>			0.9366*** (0.3112)	0.9181*** (0.3111)
<i>FirmAge</i>	−0.1369 (0.5681)	−0.1114 (0.5679)	−0.6401 (0.5779)	−0.6185 (0.5777)
<i>Size</i>	0.4833*** (0.1114)	0.4868*** (0.1114)	0.4850*** (0.1116)	0.4885*** (0.1116)
<i>Lev</i>	0.7848* (0.4403)	0.8236* (0.4413)	0.1740 (0.4488)	0.2049 (0.4495)
<i>Fixed</i>	−1.8547*** (0.5843)	−1.8287*** (0.5847)	−2.9052*** (0.6140)	−2.8927*** (0.6142)
<i>Agent</i>	−0.0215 (0.0198)	−0.0217 (0.0198)	−0.0210 (0.0198)	−0.0212 (0.0198)
<i>Indep</i>	1.0173 (1.1822)	1.0135 (1.1827)	0.8911 (1.1825)	0.8847 (1.1831)
<i>Dual</i>	−0.0063 (0.1427)	−0.0082 (0.1426)	0.0109 (0.1426)	0.0091 (0.1425)
<i>Big4</i>	−0.1083 (0.3531)	−0.1125 (0.3522)	−0.1217 (0.3545)	−0.1257 (0.3535)
<i>Firm fixed effect</i>	YES	YES	YES	YES
<i>Year fixed effect</i>	YES	YES	YES	YES
<i>Industry fixed effect</i>	YES	YES	YES	YES
<i>Constant</i>	−5.9129** (2.8052)	−6.3915** (3.1200)	−3.4815 (2.8402)	−4.0496 (3.1518)
<i>N</i>	47,509	47,509	47,509	47,509
<i>R² adjusted</i>	0.0995	0.0995	0.1003	0.1004

Note: ***, **, and * represent significance levels of 1%, 5%, and 10% respectively. Values in parentheses are standard errors.

treatment and control groups. Post-matching, the density curves align closely across the propensity score range, indicating that systematic covariate differences were substantially reduced. This improved balance satisfies the comparability assumption and validates the PSM strategy. Regression results using the matched sample (Table 7) suggesting that the main findings remain robust after addressing selection bias.

4.3.4. Placebo test

To address concerns regarding unobserved confounding variables, we conduct a placebo test by randomizing the assignment of AI adoption. We first construct a dummy variable (*AI_dummy*) and ensure its distribution in the randomized sample matches the original. Following 1000 permutations, Fig. 6 shows that the t-statistics for the pseudo-coefficients follow a normal distribution centered around zero. This indicates that the observed treatment effect is not driven by random chance or omitted factors, further validating the robustness of our primary findings.

4.3.5. GMM dynamic panel analysis

To account for serial correlation and persistence in firm-level M&A behavior, we employ a system GMM regression. As shown in Table 8, the test statistics confirm the validity of our instruments, satisfying the required GMM conditions. After controlling for the lagged dependent variable (*L.M&A*) and its potential endogeneity, the coefficient for *AI* remains positive and statistically significant at the 5% level. This further suggests that the positive association between *AI* adoption and M&A behavior is robust to the dynamic nature of corporate investment decisions.

5. Robustness tests

5.1. Changing the measurement of the independent variable

To further verify robustness, we remeasure the independent variable focusing exclusively on the Management Discussion and Analysis (MD&A) section of annual reports, following Yao et al. (2024). Since firms disclose critical strategic directions and R&D investments in this section, it provides a more targeted reflection of AI activities. We define *AI_MD&A* as the natural logarithm of one plus the count of AI-related keywords within the MD&A. As shown in Table 9, the regression results remain consistent with our primary findings, suggesting that the association between *AI* and M&A behavior is robust to this alternative variable specification.

Additionally, to ensure we capture genuine financial commitments to AI, we remeasure the independent variable using AI investment level, with data sourced from the CSMAR database. We aggregate AI-related intangible assets (e.g., software, cloud computing) and fixed assets (e.g., servers, chips, network hardware) identified through keyword filtering. The total AI investment is then scaled by the firm's total assets to derive the investment level (*AIlevel*). As shown in Table 10, the regression results remain consistent with our baseline findings, providing further evidence of the robust relationship between AI technology adoption and M&A behavior.

5.2. Changing the measurement of the dependent variable

To verify the robustness of our findings, we replace the M&A measurement with the asinh transformation of one plus the transaction value of the firm's M&A activity (*M&A_1*). The results align with the benchmark regressions, reinforcing the validity of all three hypotheses and

Table 12
Changing the measurement of the moderator variable.

	Model 24	Model 25	Model 26	Model 27
	DV = M&A	DV = M&A	DV = M&A	DV = M&A
<i>AI</i>	0.1606** (0.0714)	0.2004*** (0.0743)	0.1474** (0.0718)	0.1876** (0.0745)
<i>Market</i>		0.0326 (0.1348)		0.0447 (0.1350)
<i>AI × Market</i>		−0.0824** (0.0418)		−0.0847** (0.0418)
<i>Cashhold1</i>			−2.1891*** (0.4516)	−2.2343*** (0.4514)
<i>AI × Cashhold1</i>			1.0383*** (0.3032)	1.0101*** (0.3032)
<i>FirmAge</i>	−0.1305 (0.5477)	−0.1058 (0.5475)	−0.5493 (0.5563)	−0.5274 (0.5560)
<i>Size</i>	0.4689*** (0.1075)	0.4722*** (0.1075)	0.4680*** (0.1077)	0.4712*** (0.1077)
<i>Lev</i>	0.7614* (0.4248)	0.7990* (0.4258)	0.1492 (0.4377)	0.1781 (0.4384)
<i>Fixed</i>	−1.7926*** (0.5639)	−1.7675*** (0.5642)	−2.6308*** (0.5862)	−2.6153*** (0.5864)
<i>Agent</i>	−0.0207 (0.0192)	−0.0210 (0.0192)	−0.0208 (0.0192)	−0.0210 (0.0192)
<i>Indep</i>	0.9828 (1.1406)	0.9791 (1.1411)	0.9198 (1.1409)	0.9147 (1.1414)
<i>Dual</i>	−0.0069 (0.1376)	−0.0087 (0.1375)	0.0021 (0.1375)	0.0004 (0.1375)
<i>Big4</i>	−0.1041 (0.3410)	−0.1081 (0.3401)	−0.1380 (0.3431)	−0.1416 (0.3422)
<i>Firm fixed effect</i>	YES	YES	YES	YES
<i>Year fixed effect</i>	YES	YES	YES	YES
<i>Industry fixed effect</i>	YES	YES	YES	YES
<i>Constant</i>	−5.7715** (2.7059)	−6.2392** (3.0098)	−3.7106 (2.7327)	−4.2679 (3.0333)
<i>N</i>	47,509	47,509	47,508	47,508
<i>R²_{adjusted}</i>	0.0995	0.0996	0.1003	0.1003

Note: ***, **, and * represent significance levels of 1%, 5%, and 10% respectively. Values in parentheses are standard errors.

Table 13
Changing the sample period.

	Model 28	Model 29	Model 30	Model 31
	DV = M&A	DV = M&A	DV = M&A	DV = M&A
<i>AI</i>	0.2922*** (0.0851)	0.3221*** (0.0875)	0.2627*** (0.0858)	0.2938*** (0.0880)
<i>Market</i>		−0.0261 (0.1420)		−0.0121 (0.1419)
<i>AI × Market</i>		−0.0702 (0.0475)		−0.0745 (0.0476)
<i>Cashhold</i>			−2.7225*** (0.5328)	−2.7635*** (0.5340)
<i>AI × Cashhold</i>			1.1503*** (0.3816)	1.1229*** (0.3829)
<i>FirmAge</i>	−0.0952 (0.5886)	−0.0760 (0.5888)	−0.6994 (0.5989)	−0.6816 (0.5991)
<i>Size</i>	0.5148*** (0.1195)	0.5184*** (0.1196)	0.5168*** (0.1198)	0.5203*** (0.1199)
<i>Lev</i>	0.9077* (0.4678)	0.9415** (0.4683)	0.2078 (0.4813)	0.2358 (0.4815)
<i>Fixed</i>	−2.2663*** (0.6189)	−2.2471*** (0.6196)	−3.4743*** (0.6569)	−3.4633*** (0.6573)
<i>Agent</i>	−0.0231 (0.0244)	−0.0234 (0.0244)	−0.0222 (0.0243)	−0.0224 (0.0244)
<i>Indep</i>	1.0162 (1.3395)	1.0133 (1.3399)	0.8666 (1.3371)	0.8620 (1.3375)
<i>Dual</i>	0.1133 (0.1651)	0.1114 (0.1650)	0.1343 (0.1648)	0.1325 (0.1647)
<i>Big4</i>	−0.1392 (0.3934)	−0.1437 (0.3926)	−0.1446 (0.3942)	−0.1485 (0.3933)
<i>AI</i>		−0.0261 (0.1420)		−0.0121 (0.1419)
<i>Firm fixed effect</i>	YES	YES	YES	YES
<i>Year fixed effect</i>	YES	YES	YES	YES
<i>Industry fixed effect</i>	YES	YES	YES	YES
<i>Constant</i>	−6.4977** (2.9503)	−6.3902** (3.2503)	−3.6770 (2.9937)	−3.6797 (3.2914)
<i>N</i>	35,099	35,099	35,099	35,099
<i>R²_{adjusted}</i>	0.0967	0.0967	0.0979	0.0979

Note: ***, **, and * represent significance levels of 1%, 5%, and 10% respectively. Values in parentheses are standard errors.

Table 14
Changing the fixed effects specification(1).

	Model 32	Model 33	Model 34	Model 35
	DV = M&A	DV = M&A	DV = M&A	DV = M&A
<i>AI</i>	0.1750*** (0.0416)	0.2047*** (0.0450)	0.1773*** (0.0426)	0.2075*** (0.0457)
<i>Market</i>		−0.1318 (0.1058)		−0.1207 (0.1061)
<i>AI × Market</i>		−0.0543** (0.0273)		−0.0563** (0.0273)
<i>Cashhold</i>			−2.9422*** (0.3378)	−2.9417*** (0.3377)
<i>AI × Cashhold</i>			0.2846 (0.2129)	0.3029 (0.2127)
<i>FirmAge</i>	−0.2876* (0.1677)	−0.2974* (0.1676)	−0.3811** (0.1676)	−0.3908** (0.1675)
<i>Size</i>	0.5113*** (0.0517)	0.5121*** (0.0516)	0.5092*** (0.0512)	0.5101*** (0.0512)
<i>Lev</i>	1.3621*** (0.2612)	1.3713*** (0.2611)	0.6666** (0.2733)	0.6760** (0.2732)
<i>Fixed</i>	−1.4250*** (0.3516)	−1.3990*** (0.3526)	−2.2320*** (0.3730)	−2.2060*** (0.3737)
<i>Agent</i>	−0.0387** (0.0172)	−0.0387** (0.0172)	−0.0377** (0.0171)	−0.0377** (0.0171)
<i>Indep</i>	0.3839 (0.8658)	0.3813 (0.8650)	0.2688 (0.8659)	0.2671 (0.8652)
<i>Dual</i>	0.0739 (0.0964)	0.0746 (0.0963)	0.0895 (0.0960)	0.0902 (0.0959)
<i>Big4</i>	−0.8021*** (0.2465)	−0.7997*** (0.2463)	−0.8019*** (0.2457)	−0.7997*** (0.2455)
<i>Year fixed effect</i>	YES	YES	YES	YES
<i>Province fixed effect</i>	YES	YES	YES	YES
<i>Constant</i>	−6.3388*** (1.1811)	−5.0808*** (1.5636)	−4.9606*** (1.1860)	−3.8104** (1.5687)
<i>N</i>	47,486	47,486	47,486	47,486
<i>R² adjusted</i>	0.0501	0.0502	0.0519	0.0520

Note: ***, **, and * represent significance levels of 1%, 5%, and 10% respectively. Values in parentheses are standard errors.

Table 15
Changing the fixed effects specification(2).

	Model 36	Model 37	Model 38	Model 39
	DV = M&A	DV = M&A	DV = M&A	DV = M&A
<i>AI</i>	0.1535** (0.0714)	0.1932*** (0.0741)	0.1287* (0.0720)	0.1708** (0.0745)
<i>Market</i>		0.0232 (0.1352)		0.0345 (0.1353)
<i>AI × Market</i>		−0.0820** (0.0416)		−0.0882** (0.0417)
<i>Cashhold</i>			−2.4714*** (0.4444)	−2.5189*** (0.4447)
<i>AI × Cashhold</i>			0.8694*** (0.2981)	0.8514*** (0.2983)
<i>FirmAge</i> (0.5489)	−0.1877 (0.5488)	−0.1635 (0.5588)	−0.6783 (0.5586)	−0.6575
<i>Size</i>	0.4657*** (0.1068)	0.4692*** (0.1067)	0.4668*** (0.1068)	0.4704*** (0.1068)
<i>Lev</i>	0.8003* (0.4245)	0.8390** (0.4256)	0.1994 (0.4331)	0.2309 (0.4339)
<i>Fixed</i>	−1.7916*** (0.5620)	−1.7659*** (0.5623)	−2.8263*** (0.5907)	−2.8129*** (0.5908)
<i>Agent</i>	−0.0206 (0.0194)	−0.0208 (0.0194)	−0.0201 (0.0194)	−0.0203 (0.0194)
<i>Indep</i>	0.9557 (1.1394)	0.9501 (1.1399)	0.8363 (1.1396)	0.8281 (1.1401)
<i>Dual</i>	0.0003 (0.1375)	−0.0017 (0.1374)	0.0171 (0.1374)	0.0153 (0.1373)
<i>Big4</i>	−0.1308 (0.3428)	−0.1352 (0.3419)	−0.1432 (0.3442)	−0.1475 (0.3432)
<i>Firm fixed effect</i>	YES	YES	YES	YES
<i>Year fixed effect</i>	YES	YES	YES	YES
<i>Industry fixed effect</i>	YES	YES	YES	YES
<i>Province fixed effect</i>	YES	YES	YES	YES
<i>Constant</i>	−5.5330** (2.7045)	−5.9134** (3.0078)	−3.1383 (2.7373)	−3.5999 (3.0379)
<i>N</i>	47,486	47,486	47,486	47,486
<i>R² adjusted</i>	0.0995	0.0996	0.1004	0.1005

Note: ***, **, and * represent significance levels of 1%, 5%, and 10% respectively. Values in parentheses are standard errors.

Table 16
Double machine learning.

	Model 40	Model 41
	DV = M&A, Partial Linear Model	DV = M&A, Interactive Model
<i>AI</i>	0.1694*** (0.0325)	0.6352*** (0.0897)
<i>Constant</i>	−0.001 (0.0365)	−
<i>Controls</i>	YES	YES
<i>Year fixed effect</i>	YES	YES
<i>N</i>	47,509	47,509

Note: ***, **, and * represent significance levels of 1%, 5%, and 10% respectively. Double Machine Learning does not report the coefficients of control variables. For computational speed, we did not control for industry and firm fixed effects. Values in parentheses are standard errors.

affirming the stability of the main findings (see Table 11).

5.3. Changing the measurement of the moderator variable

To further verify robustness, we recalculate cash holdings using a broader definition. Following standard practice, we define *Cashhold1* as the ratio of cash and cash equivalents (including cash on hand and bank deposits from the cash flow statement) to total assets. As shown in Table 12, the regression results remain consistent, providing additional support for our research hypotheses.

5.4. Changing the sample period

Table 13 reports regression results excluding the 2020–2022 period

to account for pandemic-related distortions. The coefficients for *AI* remain significantly positive, reaffirming H1. However, the interaction term *AI×Market* loses statistical significance, likely due to post-pandemic policy uncertainties that weakened regional institutional advantages. Conversely, the interaction *AI×Cashhold* remains significantly positive (Models 30 and 31), further validating H3. This suggests that firms with higher liquidity are better positioned to leverage *AI* in supporting their M&A strategies, even when controlling for exogenous shocks.

5.5. Changing the fixed effects specification

To further verify robustness, we re-estimate the model by including province and year fixed effects. The results are highly consistent with the baseline estimates, suggesting that our findings remain stable under alternative fixed-effect structures (see Table 14).

To enhance robustness, we incorporate province fixed effects alongside the original firm, year, and industry fixed effects to account for time-invariant regional heterogeneity. As shown in Models 36–39 (Table 15), the coefficients for *AI* remain significantly positive, and the moderating effects stay statistically significant. These results confirm that the main findings are stable under more stringent fixed-effect specifications.

5.6. Double machine learning

To address potential bias from high-dimensional covariates, we conduct a robustness test using Double Machine Learning (DML), following Chernozhukov et al. (2018); Zhou, Liu, et al. (2025b); and Wu et al. (2025). DML provides reliable causal estimates by using machine

Table 17
Changing the clustering specification.

	Model 42	Model 43	Model 44	Model 45	Model 46	Model 47	Model 48	Model 49	Model 50
	DV = M&A	DV = M&A	DV = M&A	DV = M&A	DV = M&A	DV = M&A	DV = M&A	DV = M&A	DV = M&A
<i>AI</i>	0.2004*** (0.0689)	0.1372** (0.0660)	0.1793*** (0.0690)	0.2004*** (0.0770)	0.1372* (0.0748)	0.1793** (0.0774)	0.2006** (0.0781)	0.1374* (0.0758)	0.1794** (0.0785)
<i>Market</i>	0.0326 (0.1223)		0.0445 (0.1223)	0.0326 (0.1399)		0.0445 (0.1401)	0.0326 (0.1421)		0.0445 (0.1422)
<i>AI × Market</i>	−0.0824** (0.0381)		−0.0884** (0.0381)	−0.0824* (0.0430)		−0.0884** (0.0430)	−0.0823* (0.0439)		−0.0884** (0.0439)
<i>Cashhold</i>		−2.4082*** (0.4167)	−2.4572*** (0.4170)		−2.4082*** (0.4619)	−2.4572*** (0.4621)		−2.4151*** (0.4674)	−2.4644*** (0.4677)
<i>AI × Cashhold</i>		0.9024*** (0.2757)	0.8846*** (0.2756)		0.9024*** (0.3091)	0.8846*** (0.3093)		0.8955*** (0.3135)	0.8775*** (0.3138)
<i>FirmAge</i>	−0.1058 (0.5105)	−0.6161 (0.5152)	−0.5951 (0.5152)	−0.1058 (0.5701)	−0.6161 (0.5795)	−0.5951 (0.5793)	−0.1056 (0.5767)	−0.6161 (0.5871)	−0.5952 (0.5869)
<i>Size</i>	0.4722*** (0.0994)	0.4705*** (0.0995)	0.4739*** (0.0995)	0.4722*** (0.1093)	0.4705*** (0.1095)	0.4739*** (0.1094)	0.4720*** (0.1118)	0.4701*** (0.1119)	0.4735*** (0.1119)
<i>Lev</i>	0.7990** (0.3925)	0.1720 (0.4029)	0.2018 (0.4033)	0.7990* (0.4347)	0.1720 (0.4434)	0.2018 (0.4441)	0.7992* (0.4471)	0.1711 (0.4553)	0.2009 (0.4561)
<i>Fixed</i>	−1.7675*** (0.5276)	−2.8063*** (0.5526)	−2.7942*** (0.5526)	−1.7675*** (0.5854)	−2.8063*** (0.6146)	−2.7942*** (0.6148)	−1.7697*** (0.5924)	−2.8103*** (0.6218)	−2.7982*** (0.6220)
<i>Agent</i>	−0.0210 (0.0195)	−0.0203 (0.0196)	−0.0205 (0.0195)	−0.0210 (0.0198)	−0.0203 (0.0198)	−0.0205 (0.0198)	−0.0210 (0.0204)	−0.0203 (0.0204)	−0.0205 (0.0204)
<i>Indep</i>	0.9791 (1.1162)	0.8609 (1.1160)	0.8547 (1.1159)	0.9791 (1.1824)	0.8609 (1.1829)	0.8547 (1.1834)	0.9491 (1.2035)	0.8299 (1.2033)	0.8242 (1.2038)
<i>Dual</i>	−0.0087 (0.1308)	0.0097 (0.1307)	0.0080 (0.1307)	−0.0087 (0.1430)	0.0097 (0.1430)	0.0080 (0.1429)	−0.0073 (0.1448)	0.0112 (0.1447)	0.0095 (0.1446)
<i>Big4</i>	−0.1081 (0.3415)	−0.1170 (0.3417)	−0.1209 (0.3416)	−0.1081 (0.3535)	−0.1170 (0.3559)	−0.1209 (0.3550)	−0.1080 (0.3589)	−0.1167 (0.3612)	−0.1206 (0.3602)
<i>Firm fixed effect</i>	YES	YES	YES	YES	YES	YES	YES	YES	YES
<i>Year fixed effect</i>	YES	YES	YES	YES	YES	YES	YES	YES	YES
<i>Industry fixed effect</i>	YES	YES	YES	YES	YES	YES	YES	YES	YES
<i>Firm – year clustering</i>	YES	YES	YES	NO	NO	NO	NO	NO	NO
<i>Firm – industry clustering</i>	NO	NO	NO	YES	YES	YES	NO	NO	NO
<i>Firm – province clustering</i>	NO	NO	NO	NO	NO	NO	YES	YES	YES
<i>Constant</i>	−6.2392** (2.8091)	−3.4251 (2.5679)	−3.9791 (2.8266)	−6.2392** (3.1126)	−3.4251 (2.8301)	−3.9791 (3.1475)	−6.2200** (3.1600)	−3.4003 (2.8744)	−3.9549 (3.1928)
<i>N</i>	47,509	47,509	47,509	47,509	47,509	47,509	47,486	47,486	47,486
<i>R² adjusted</i>	0.0996	0.1004	0.1005	0.0996	0.1004	0.1005	0.0996	0.1004	0.1005

Note: ***, **, and * represent significance levels of 1%, 5%, and 10% respectively. Values in parentheses are standard errors.

Table 18
Changing econometric models.

	Model 51	Model 52	Model 53	Model 54
	DV = M&A	DV = M&A	DV = M&A	DV = M&A
AI	0.0343** (0.0148)	0.0408*** (0.0151)	0.0305** (0.0150)	0.0373** (0.0153)
Market		−0.0047 (0.0252)		−0.0015 (0.0253)
AI × Market		−0.0153* (0.0088)		−0.0164* (0.0089)
Cashhold			−0.5452*** (0.1006)	−0.5551*** (0.1006)
AI × Cashhold			0.1789*** (0.0687)	0.1718** (0.0691)
FirmAge	−0.0234 (0.1093)	−0.0178 (0.1094)	−0.1352 (0.1135)	−0.1297 (0.1136)
Size	0.1361*** (0.0224)	0.1365*** (0.0224)	0.1362*** (0.0225)	0.1365*** (0.0225)
Lev	0.1657* (0.0877)	0.1742** (0.0878)	0.0301 (0.0905)	0.0373 (0.0906)
Fixed	−0.3076** (0.1210)	−0.3058** (0.1209)	−0.5190*** (0.1262)	−0.5190*** (0.1260)
Agent	−0.0043 (0.0039)	−0.0044 (0.0039)	−0.0041 (0.0039)	−0.0042 (0.0039)
Indep	0.2208 (0.2338)	0.2208 (0.2338)	0.1993 (0.2340)	0.1990 (0.2340)
Dual	0.0010 (0.0286)	−0.0000 (0.0286)	0.0045 (0.0286)	0.0035 (0.0286)
Big4	−0.0019 (0.0789)	−0.0031 (0.0787)	−0.0072 (0.0792)	−0.0083 (0.0790)
Firm fixed effect	YES	YES	YES	YES
Year fixed effect	YES	YES	YES	YES
Industry fixed effect	YES	YES	YES	YES
Constant	−1.2070** (0.5589)	−1.1887* (0.6078)	−0.6787 (0.5690)	−0.6846 (0.6166)
N	40,705	40,705	40,705	40,705

Note: ***, **, and * represent significance levels of 1%, 5%, and 10% respectively. Values in parentheses are standard errors.

learning to flexibly control for confounding variables. We implement both partial linear and interactive models estimated via random forest, incorporating firm and year fixed effects alongside primary controls. To

ensure stability and generalizability, we employ 5-fold cross-validation throughout the estimation. The results in Table 16 confirm that our findings remain robust under these advanced causal inference specifications.

Table 19
Heterogeneity analysis based on ownership.

	Model 55	Model 56
	DV = M&A, SOE	DV = M&A, non – SOE
AI	0.0211 (0.1351)	0.2438*** (0.0827)
FirmAge	1.2133 (0.8629)	−0.4056 (0.7123)
Size	0.4702*** (0.1746)	0.5636*** (0.1385)
Lev	−0.3318 (0.6747)	2.0263*** (0.5503)
Fixed	−1.2080 (0.8652)	−2.1046*** (0.7390)
Agent	−0.0340 (0.0248)	0.0060 (0.0292)
Indep	1.3966 (1.7083)	−0.1095 (1.5202)
Dual	0.6694** (0.2789)	−0.2571* (0.1558)
Big4	0.0171 (0.4489)	−0.1952 (0.5073)
Firm fixed effect	YES	YES
Year fixed effect	YES	YES
Industry fixed effect	YES	YES
Constant	−9.2959** (4.4896)	−7.1635** (3.4521)
N	17,497	30,011
R ² adjusted	0.0895	0.1128

Note: ***, **, and * represent significance levels of 1%, 5%, and 10% respectively. Values in parentheses are standard errors.

5.7. Changing the clustering specification

To further test the robustness of our inferences, we re-estimate the benchmark models using alternative clustering specifications for standard errors, including firm-year, firm-industry, and firm-province levels. The results (Table 17) remain highly consistent with the baseline regressions, indicating that our findings are not driven by specific assumptions regarding the clustering of error terms.

5.8. Changing econometric models

Linear regression is limited here due to the large number of zeros in the dependent variable. Following Cohn et al. (2022), we employ pseudo-Poisson regression to address potential OLS bias; the results remain consistent (Table 18).

6. Heterogeneity analysis

6.1. Ownership heterogeneity

Table 19 reveals significant heterogeneity across ownership types. For state-owned enterprises (SOEs), the coefficient for AI is small and statistically insignificant ($\beta = 0.0211$, $p > 0.1$). Conversely, for non-SOEs, the coefficient is positive and significant at the 1% level ($\beta = 0.2438$, $p < 0.01$), indicating that AI adoption substantially promotes M&A activity in this group.

This divergence aligns with the theory of relational versus

Table 20
Heterogeneity analysis based on industry.

	Model 57 DV = M&A, Regulated	Model 58 DV = M&A, Non – regulated	Model 59 DV = M&A, Non – service	Model 60 DV = M&A, Service
<i>AI</i>	0.0937 (0.1319)	0.1940** (0.0870)	0.2427*** (0.0829)	−0.0634 (0.1532)
<i>FirmAge</i>	−0.0697 (0.9441)	−0.5214 (0.7238)	−0.4018 (0.6177)	1.0311 (1.4584)
<i>Size</i>	0.7286*** (0.1908)	0.4187*** (0.1411)	0.4296*** (0.1214)	0.8274*** (0.2747)
<i>Lev</i>	−0.4742 (0.7458)	2.1291*** (0.5382)	1.2621*** (0.4655)	0.9561 (1.0987)
<i>Fixed</i>	−2.8959*** (0.8972)	−0.9062 (0.7483)	−2.1845*** (0.6098)	0.8075 (1.7862)
<i>Agent</i>	−0.0338 (0.0324)	−0.0078 (0.0245)	−0.0219 (0.0231)	−0.0063 (0.0337)
<i>Indep</i>	0.0354 (1.9687)	1.4127 (1.4118)	1.0578 (1.2537)	1.0191 (2.8084)
<i>Dual</i>	−0.1903 (0.2449)	0.1397 (0.1667)	−0.0624 (0.1528)	0.2560 (0.3266)
<i>Big4</i>	−0.9369 (0.6164)	0.3268 (0.4178)	−0.0089 (0.3953)	0.0250 (0.7557)
<i>Firm fixed effect</i>	YES	YES	YES	YES
<i>Year fixed effect</i>	YES	YES	YES	YES
<i>Industry fixed effect</i>	YES	YES	YES	YES
<i>Constant</i>	−10.0606** (4.7499)	−4.7252 (3.5356)	−4.2873 (3.0487)	−17.2783** (7.1189)
<i>N</i>	16,731	30,743	38,954	8509
<i>R²_adjusted</i>	0.1052	0.0991	0.1015	0.1119

Note: ***, **, and * represent significance levels of 1%, 5%, and 10% respectively. Values in parentheses are standard errors.

Table 21
Heterogeneity analysis based on analyst attention.

	Model 61 DV = M&A, High – Attention	Model 62 DV = M&A, Low – Attention
<i>AI</i>	0.1899 (0.1162)	0.2054** (0.0983)
<i>FirmAge</i>	−0.5423 (0.8029)	−0.0588 (0.8586)
<i>Size</i>	−0.1345 (0.2019)	0.5169*** (0.1544)
<i>Lev</i>	3.5720*** (0.8019)	−0.4758 (0.5482)
<i>Fixed</i>	−2.0237** (1.0127)	−1.2300* (0.7091)
<i>Agent</i>	−0.0110 (0.0282)	−0.0254 (0.0253)
<i>Indep</i>	0.1880 (1.8480)	1.5943 (1.5365)
<i>Dual</i>	−0.0947 (0.2242)	0.0161 (0.1806)
<i>Big4</i>	−0.2062 (0.4340)	0.2958 (0.5981)
<i>Firm fixed effect</i>	YES	YES
<i>Year fixed effect</i>	YES	YES
<i>Industry fixed effect</i>	YES	YES
<i>Constant</i>	8.9005* (4.9052)	−7.4217* (4.0881)
<i>N</i>	21,698	24,816
<i>R²_adjusted</i>	0.1184	0.0843

Note: ***, **, and * represent significance levels of 1%, 5%, and 10% respectively. Values in parentheses are standard errors.

transactional ownership (Aguilera & Crespi-Cladera, 2016; Mariotti et al., 2023). SOEs, as relational owners, prioritize long-term social and political objectives over rapid technological shifts, leading to greater caution in leveraging AI for strategic expansion. In contrast, non-SOEs—characterized by transactional ownership—prioritize market efficiency. Their flexible governance allows them to utilize AI to reduce informational frictions and optimize resource allocation, thereby enhancing their strategic capacity for M&A.

Table 22
Heterogeneity analysis based on firms' geographical location.

	Model 63 DV = M&A, Western Region	Model 64 DV = M&A, Central Region	Model 65 DV = M&A, Eastern Region
<i>AI</i>	0.1018 (0.2119)	0.0104 (0.1956)	0.2176*** (0.0821)
<i>FirmAge</i>	1.4412 (1.5889)	3.9047*** (1.3823)	−0.9192 (0.6248)
<i>Size</i>	0.7602*** (0.2772)	0.1890 (0.2590)	0.4941*** (0.1301)
<i>Lev</i>	0.6908 (1.0883)	0.5198 (1.1045)	0.9689* (0.5058)
<i>Fixed</i>	−0.1561 (1.3509)	−1.9737 (1.3814)	−2.0845*** (0.6938)
<i>Agent</i>	−0.0101 (0.0393)	−0.0216 (0.0530)	−0.0249 (0.0236)
<i>Indep</i>	−0.1980 (3.1288)	3.0923 (2.7879)	0.5611 (1.3612)
<i>Dual</i>	0.1888 (0.4785)	0.1325 (0.3719)	−0.0600 (0.1545)
<i>Big4</i>	0.2420 (0.8993)	0.6713 (0.8998)	−0.3018 (0.3995)
<i>Firm fixed effect</i>	YES	YES	YES
<i>Year fixed effect</i>	YES	YES	YES
<i>Industry fixed effect</i>	YES	YES	YES
<i>Constant</i>	−16.5834** (7.7587)	−11.5889* (6.6824)	−4.0235 (3.1687)
<i>N</i>	5302	7940	34,235
<i>R²_adjusted</i>	0.0731	0.0887	0.1073

Note: ***, **, and * represent significance levels of 1%, 5%, and 10% respectively. Values in parentheses are standard errors.

6.2. Industry heterogeneity

Table 20 reveals how industry characteristics moderate the association between AI and M&A behavior across two dimensions: (1) Regulated vs. Non-Regulated Industries. The results show a muted effect in regulated industries but a significant positive effect in non-regulated

Table 23
Economic consequences test.

	Model 66	Model 67
	DV = TobinQ	DV = F.TobinQ
M&A	0.0060*** (0.0006)	0.0011* (0.0006)
FirmAge	0.9828*** (0.0926)	0.5986*** (0.0856)
Size	−0.5420*** (0.0257)	−0.5306*** (0.0250)
Lev	0.4576*** (0.0824)	0.4467*** (0.0817)
Fixed	0.0641 (0.1023)	0.1379 (0.1004)
Agent	0.0010 (0.0034)	0.0019 (0.0022)
Indep	0.4502** (0.1820)	0.2854 (0.1827)
Dual	−0.0477** (0.0200)	−0.0258 (0.0206)
Big4	0.0446 (0.0566)	0.0276 (0.0659)
Firm fixed effect	YES	YES
Year fixed effect	YES	YES
Industry fixed effect	YES	YES
Constant	10.7412*** (0.5985)	11.6987*** (0.5805)
N	47,509	41,186
R ² _adjusted	0.5485	0.5895

Note: ***, **, and * represent significance levels of 1%, 5%, and 10% respectively. Values in parentheses are standard errors.

sectors (significant at the 5% level). This divergence stems from institutional flexibility. Regulated industries face strict oversight and administrative constraints that inhibit managerial autonomy and delay AI integration. Conversely, non-regulated industries benefit from market-oriented governance, allowing firms to leverage AI more effectively to mitigate information asymmetries and identify synergistic targets. (2) Service vs. Non-Service Industries. Significant heterogeneity also exists between sectors. In non-service industries, AI adoption is positively associated with ($\beta = 0.2427, p < 0.01$), as AI's focus on operational efficiency drives M&A involving tangible assets. In contrast, for service industries, the effect is statistically insignificant ($\beta = -0.0634$). This suggests that AI in services primarily enhances internal informational advantages and intangible value, rather than directly driving external M&A activity.

6.3. Analyst attention heterogeneity

Table 21 presents the heterogeneity analysis based on analyst attention (proxied by the log of analyst followers). The results reveal differentiated associations of AI adoption: For Low Analyst Coverage, the AI coefficient is significantly positive (5% level). In these less transparent environments, severe informational frictions allow AI to play a more salient role in reducing asymmetries and improving M&A decision-making efficiency. For High Analyst Coverage, the AI coefficient is statistically insignificant. Robust external monitoring and existing disclosure mechanisms in these firms already alleviate information frictions, thereby diminishing the marginal benefit of AI adoption in driving M&A behavior.

6.4. Geographic location heterogeneity

Table 22 examines the association between AI adoption and geographical regions, revealing notable regional disparities. In the Eastern region, the coefficient is significantly positive at the 1% level, suggesting that firms in economically advanced areas—supported by superior digital infrastructure and institutional environments—are better positioned to leverage AI for M&A activities.

Conversely, in the Western and Central regions, the coefficients for AI adoption are positive but insignificant. This indicates that the strategic role of AI in shaping M&A behavior is constrained in less-developed areas, likely due to weaker digital ecosystems and institutional support. These findings underscore how regional development disparities determine the effectiveness of digital transformation.

6.5. Further analysis: economic consequences

To assess the economic implications of AI-driven M&A, Table 23 examines its association with firm value (Tobin's Q). This metric captures market expectations of a firm's growth potential relative to the replacement cost of its assets. Across both Models 66 and 67, the coefficients for *Masize* are significantly positive, indicating that firms with more active M&A engagement exhibit higher market valuations.

These results suggest that M&A behavior contributes positively to forward-looking performance in the capital market. The findings imply that AI-driven M&A activities do more than just reshape organizational boundaries; they serve as a favorable signal to investors regarding a firm's strategic agility and market positioning, ultimately translating into enhanced market value.

7. Conclusions

As AI becomes increasingly integrated into firm operations (Dong et al., 2024; Hruby, 2025; Liu, Wang, et al., 2024b; Liu, Zhang, & Qi, 2025c), it exerts a profound influence on firm financial behavior, particularly M&A. Our results show that AI adoption significantly drives M&A strategies. As a critical case study, our findings within China's unique framework provide a strategic benchmark for other emerging markets undergoing digital transformations. While prior research often views digital technology as a passive acquisition target or focuses on "AI M&A"—where firms acquire AI-related assets to enhance innovation (Tang, Liang, & Fang, 2024a) or diversification (Zhou, Liang, et al., 2025a)—our study identifies internal AI adoption as a proactive ex ante determinant of acquisition propensity. This distinction shifts the focus from buying AI to using AI as a strategic enabler to execute deals. We find this catalytic effect is heterogeneous: it is strongest in firms with substantial cash holdings, suggesting that financial slack is essential to convert technological capability into strategic action. Furthermore, marketization negatively moderates this relationship; in less-marketized regions, AI serves as a critical tool for overcoming the information asymmetries that otherwise inhibit deal-making.

This study offers two main managerial implications bridging technology and finance. First, firms should recognize AI as a strategic asset for corporate control decisions. Senior management should integrate AI directly into the M&A workflow—using it for market scanning, data-driven due diligence, and forecasting postmerger synergies. Treating AI as part of the financial infrastructure enables firms to build a competitive advantage in the market for corporate control. Second, our findings on cash holdings and market conditions provide guidance for capital allocation. For cash-rich firms, investing in AI-assisted decision systems is essential for deploying capital efficiently and maximizing shareholder returns. For firms expanding into opaque or less-developed markets, AI serves as a navigation tool that mitigates risk and compensates for underdeveloped market intermediaries.

Nevertheless, we acknowledge that the generalizability of our findings may be bounded by the specific institutional context of China. As a leading emerging economy with rapid digital infrastructure growth and unique capital market characteristics, China provides a critical laboratory for observing the association between AI and its outcomes; however, the mechanisms identified here—particularly those tied to marketization levels—may manifest differently in economies with different regulatory frameworks or more mature information intermediaries. Like any study, this work has limitations that present opportunities for future research. First, different departments—such as

marketing and finance—may deploy AI in distinct ways. Although we identify an aggregate effect, we cannot determine which functional applications most strongly influence M&A outcomes. Future research could use granular operational data or case studies to explore how specific AI uses interact with human—capital reallocation, a key source of diversification synergies (Tate & Yang, 2024). Second; while our analysis focuses on listed firms in a single institutional setting; future studies could expand these boundary conditions to incorporate cross-

country comparisons. Such research could include governance characteristics (Xiao et al., 2024) or macrolevel uncertainty (Zhang et al., 2025) across diverse legal systems. Examining how AI adoption interacts with managerial risk preferences and economic volatility under different institutional logics would deepen the RBV in emerging markets and enhance understanding of the financial dynamics shaping digital—era M&A on a global scale.

Appendix A

Table A1
Artificial intelligence (AI) glossary.

AI	AI Products	AI Chip	Machine Translation	Machine Learning
Computer Vision	Human–Computer Interaction	Deep Learning	Neural Network	Biometric Recognition
Image Recognition	Data Mining	Feature Recognition	Speech Synthesis	Speech Recognition
Knowledge Graph	Smart Banking	Smart Insurance	Human–Machine Collaboration	Intelligent Supervision
Intelligent Education	Intelligent Customer Service	Smart Retail	Smart Agriculture	Intelligent Investment Advisor
Augmented Reality	Virtual Reality	Smart Healthcare	Smart Speaker	Intelligent Speech
Smart Governance	Autonomous Driving	Smart Transportation	Convolutional Neural Network	Voiceprint Recognition
Feature Extraction	Autonomous Vehicle	Smart Home	Question–Answering System	Facial Recognition
Business Intelligence	Intelligent Finance	Recurrent Neural Network	Reinforcement Learning	Intelligent Agent
Smart Elderly Care	Big Data Marketing	Big Data Risk Control	Big Data Analysis	Big Data Processing
Support Vector Machine (SVM)	Long Short–Term Memory (LSTM)	Robotic Process Automation	Natural Language Processing	Distributed Computing
Knowledge Representation	Intelligent Chip	Wearable Devices	Big Data Management	Smart Sensors
Pattern Recognition	Edge Computing	Big Data Platform	Intelligent Computing	Intelligent Search
Internet of Things	Cloud Computing	Augmented Intelligence	Voice Interaction	Intelligent Environment
Human–Computer Dialogue	Deep Neural Network	Big Data Operations		

Note: AI = Artificial Intelligence.

Data availability

Data will be made available on request from the corresponding author.

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